Disentangling Between-Target and Within-Target Signal in PPI $\Delta\Delta G$ Benchmarking

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Abstract

Predicting how mutations alter protein–protein binding affinity ($\Delta\Delta G$) underpins antibody engineering, interface optimisation and the interpretation of disease-associated variants. The pooled correlation commonly reported for such benchmarks is difficult to interpret directly, because it depends both on whether a model places different targets on the correct overall binding scale and on whether it orders mutations correctly within each target. We formulate an exact between-/within-target covariance decomposition for PPI $\Delta\Delta G$ benchmarking, which writes pooled Pearson correlation as $r = C_{\mathrm{B}} + C_{\mathrm{W}}$, and apply it to matched and provenance-filtered evaluation sets, with explicit sensitivity analyses for repeated mutation keys. In a controlled comparison that changes only the split regime (identical 7,523 mutation rows and identical model configurations), median pooled Pearson falls from 0.6844 under random splitting to 0.1508 under family-out-of-distribution evaluation, with $\Delta C_{\mathrm{B}} = -0.4347$ and $\Delta C_{\mathrm{W}} = -0.1011$; on an absolute-change basis the between-target component accounts for 81.1% of the summed absolute component change while the within-target component remains positive. Target weighting changes how these contributions should be read: over the five schemes C_{B} reverses sign in 40/109 configurations and in 34/109 even once $|C_{\mathrm{B}}| > 0.005$, and C_{W} reverses sign in 5/109; both components therefore move with the weighting convention. Fixed-K shortlist metrics are constrained by candidate-set geometry: for $K = 10$, a rank-linear discriminability surrogate $D(n, K)$ is zero when $n \leq K$ and peaks near $n^* \approx 20.5$, and 40.8% of targets with $n \geq 5$ fall in that degenerate regime. In a matched retrospective ablation the decomposition separated a change that improved pooled scores only from one that also improved within-target ranking. External evidence comprised five methods and is underpowered; its apparent cross-axis agreement is not robust to score orientation, so the panel neither supports nor refutes a universal decoupling of pooled and within-target performance. We recommend that PPI $\Delta\Delta G$ benchmarks report both components under an explicitly declared weighting, together with candidate-set geometry.

1 Introduction

Predicting how mutations alter protein–protein binding affinity ($\Delta\Delta G$) is central to antibody affinity maturation, interface optimisation and the interpretation of disease-associated variants. As the number of deep-learning methods grows rapidly, progress is increasingly judged by benchmark evaluation, and the properties of the evaluation protocol itself — what it measures, and under which extrapolation conditions — determine what the field can conclude.

Current practice relies largely on pooled metrics, computing Pearson or Spearman correlation and RMSE over the entire mutation set[13], often supplemented by per-complex or per-target metrics. Bushuiev et al.

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quantified structural leakage in PPI data: about 65% of test examples have a near-duplicate training example under PDB-code splitting, about 30% even under 30% sequence-identity splitting, 56% for SKEMPI v2.0[1] split by PPI code, and 0% under domain-expert grouping[2]. CATH-ddG adopted CATH homologous-superfamily splits and defined hard mutations by a maximum TM-score below 0.6[3]; related methods likewise report per-structure or per-target correlations[4;5;10]. We emphasise that leakage-aware splitting, family-out-of-distribution evaluation, per-target reporting and shortlist (Top-K) evaluation were none of them introduced by this work.

A key question nevertheless remains unanswered directly: where does pooled benchmark performance come from? Reporting pooled and per-target metrics side by side does not quantify the additive contribution of between-target and within-target signal to pooled performance. In benchmarks containing many targets with highly unequal mutation counts, pooled correlation is additionally dominated by large targets, which amplifies the conflation.

Here we address this question by instantiating, for PPI $\Delta\Delta G$ benchmarking, an exact between-/within-target covariance decomposition. The decomposition follows from the law of total covariance; our contribution lies in the benchmark-specific formalisation and in its systematic integration with a matched, provenance-filtered evaluation framework, not in the covariance identity itself. It makes 'what pooled performance is made of' a directly measurable quantity. We quantify four things: which component is changed by split strictness; how target weighting changes the interpretation of the two components; which constraints candidate-set geometry places on fixed-K shortlist evaluation; and whether the decomposition can distinguish model modifications that improve pooled scores only from those that also improve within-target ranking. We further bound the evidence using five independently developed external methods.

2 Results

Table 1: Dataset and evaluation overview.

Evaluation set	Rows	Targets	Targets ($n \geq 5$)	Method families	Purpose
Full annotation namespace	4,473	214	130	—	Namespace definition
Primary matched set	3,351	166	102	10	Main capability-resolved analyses
Controlled split set	7,523	230	172	12	Split-strictness experiment
External unambiguous set	1,781	168	67	5	Boundary of generalisation

All rows are joined by explicit row identifiers.

Pooled PPI $\Delta\Delta G$ performance contains separable between-target and within-target signal

For a mutation i belonging to target c, the observed $\Delta\Delta G$ can be written as a target-level mean plus a mutation-specific deviation from that mean:

$$y_{ci} = \mu_c + \varepsilon_{ci} \qquad \hat{y}_{ci} = \hat{\mu}_c + \hat{\varepsilon}_{ci} \tag{1}$$

With σ_y and $\sigma_{\hat{y}}$ the pooled standard deviations, the pooled correlation splits exactly as

$$r_{\text{pooled}} = \frac{\text{Cov}(y, \hat{y})}{\sigma_y \sigma_{\hat{y}}} = C_{\text{B}} + C_{\text{W}}, \tag{2}$$

where C_{B} and C_{W} are the between-target and within-target covariance contributions. The identity holds **exactly for every individual configuration**: across the 36 controlled-split configurations the maximum absolute residual $|r_{\text{pooled}} - C_{\text{B}} - C_{\text{W}}|$ is 4.4×10^{-16}, i.e. machine precision. The values reported in Table 2 are however *marginal medians*, computed separately for r_{pooled}, C_{B} and C_{W}; because the median is not additive, those three displayed medians need not sum exactly (e.g. Random: $0.4275 + 0.2591 = 0.6866$ versus a median r_{pooled} of 0.6844). These are covariance-signal contributions on a common scale, not independent correlations, skill scores or causal mechanisms.

We apply this decomposition on a matched evaluation set of 3,351 mutation rows and 102 targets with at least five mutations, drawn from 10 independent method families. Three distinct row universes appear in this work and are defined in Methods; the primary matched universe (3,351 rows) is a provenance- and feature-filtered subset of the 4,473-row annotation namespace, which is itself a feature-coverage subset of the 7,523-row controlled-split universe.

Family-OOD evaluation disproportionately removes between-target signal

To isolate the effect of split strictness, we compared identical model configurations on the same 7,523 mutation rows while changing only the split regime. Median pooled Pearson falls from 0.6844 to 0.1508. The decline is not uniform: $\Delta C_{\mathrm{B}} = -0.4347$ and $\Delta C_{\mathrm{W}} = -0.1011$. On an absolute-change basis the between-target component accounts for 81.1% of the *summed absolute component change*,

$$\frac{|\Delta C_{\mathrm{B}}|}{|\Delta C_{\mathrm{B}}| + |\Delta C_{\mathrm{W}}|} = 0.8113, \tag{3}$$

while C_{W} remains positive ($+0.1580$). We verified this attribution on the paired scale as well. Because the same model specifications and seeds were evaluated under both regimes, all 9 configurations are directly matched: the paired differences satisfy $\Delta r_j = \Delta C_{\mathrm{B},j} + \Delta C_{\mathrm{W},j}$ exactly (maximum residual 4.2×10^{-16}), with medians $\Delta r = -0.5332$, $\Delta C_{\mathrm{B}} = -0.4324$ and $\Delta C_{\mathrm{W}} = -0.1011$. The paired attribution is therefore 81.0% (configuration-cluster bootstrap 95% CI 71.0–84.5%), essentially identical to the marginal-median figure. This is not the same proposition as 'the model memorised target identity': a more defensible reading is that the pattern is consistent with reliance on target-specific information that does not transfer across families. The `seq40` partition is a distinct, sequence-identity-based split: after the key-level correction described in Methods it yields 301 units (versus 230 for the complex-level split), and it agrees with the complex-level assignment on only about 20% of rows. We therefore report it separately rather than treating it as a duplicate of the complex-level comparison.

Table 2: Controlled split comparison.

Split regime	Pooled Pearson	C_{B}	C_{W}	Macro Spearman	RMSE
Random	0.6844	+0.4275	+0.2591	0.0707	1.7678
Complex	0.1931	+0.0233	+0.1685	0.0497	2.6412
seq40	0.1961	+0.0239	+0.1721	0.0631	2.6911
Family-OOD	0.1508	-0.0072	+0.1580	0.0377	2.6952

Median values across configurations. `seq40` is reported separately; complex in this dataset.

Target aggregation changes the interpretation of covariance contributions

Holding predictions fixed and varying only the aggregation weights across five schemes (per-mutation, per-target, proportional to $\sqrt{n}$, proportional to $\log(1+n)$, and capped at 30), Treating a_c as the target's total weight (so that the per-target scheme genuinely equalises targets; Supplementary Note 5), C_{B} is directionally unstable: 40/109 configurations reverse sign at some magnitude, 36/109 once $|C_{\mathrm{B}}| > 0.001$ and 34/109 once $|C_{\mathrm{B}}| > 0.005$, so the instability is not merely a near-zero artefact. C_{W} is markedly more stable — it reverses sign in only 5/109 configurations, and in 4/109 once $|C_{\mathrm{W}}| > 0.005$ — but neither component is invariant: their median cross-scheme ranges are 0.3052 for C_{B} and 0.1240 for C_{W}. Both therefore move with the weighting convention, which is why any reported decomposition should declare its weighting scheme. Under the alternative row-multiplier reading the same data give 0.0930 and 0.0624, with C_{W} reversing sign in 0/109; the qualitative contrast between the two components is preserved under either reading (Supplementary Note 5).

Figure 1 | Capability-resolved evaluation of PPI ΔΔG predictors

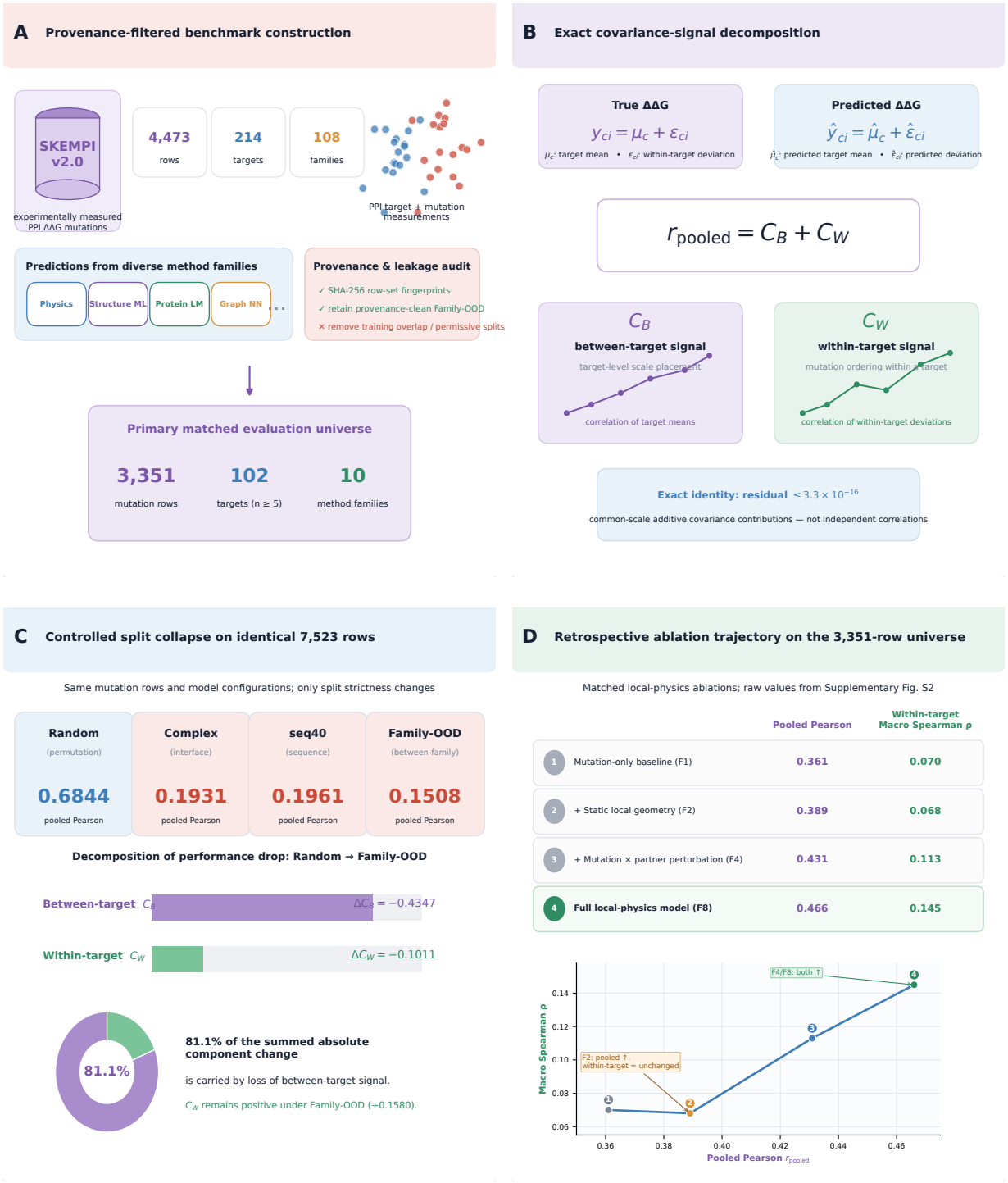


Figure 1: Capability-resolved evaluation of PPI $\Delta\Delta G$ predictors. (A) Provenance-filtered, fingerprint-matched benchmark construction: the SKEMPI v2.0 annotation namespace (4,473 rows, 214 targets, 108 *structural* families — distinct from the 10 *method* families) is audited by SHA-256 row-set fingerprinting; predictions whose recorded training split is permissive, or whose evaluation rows may overlap their training folds, are excluded, giving a primary matched universe of 3,351 rows, 102 targets with $n \geq 5$ and 10 method families. The annotation namespace is itself a feature-coverage subset of the 7,523-row controlled-split universe (Methods, *Dataset universes*). (B) Exact additive covariance-signal decomposition. The identity $r_{\text{pooled}} = C_{\text{B}} + C_{\text{W}}$ holds for every individual configuration (maximum residual 4.4×10^{-16}); the two components are common-scale additive contributions, not independent correlations or causal mechanisms. (C) Controlled split collapse on the identical 7,523-row universe with identical model specifications and seeds; only the split regime changes. Pooled correlation falls from 0.6844 to 0.1508, and 81.1% of the summed absolute component change is carried by the between-target component. (D) Retrospective matched ablation on the identical 3,351-row evaluation universe. Pooled Pearson and within-target Macro Spearman are shown for the mutation-only baseline (F1), static local geometry (F2), mutation × partner perturbation (F4), and the full local-physics model (F8). Static local geometry improves pooled correlation while leaving within-target ranking essentially unchanged, whereas F4 and F8 improve both axes. The capability trajectory is retrospective and is not intended as a causal or prospective model-selection analysis.

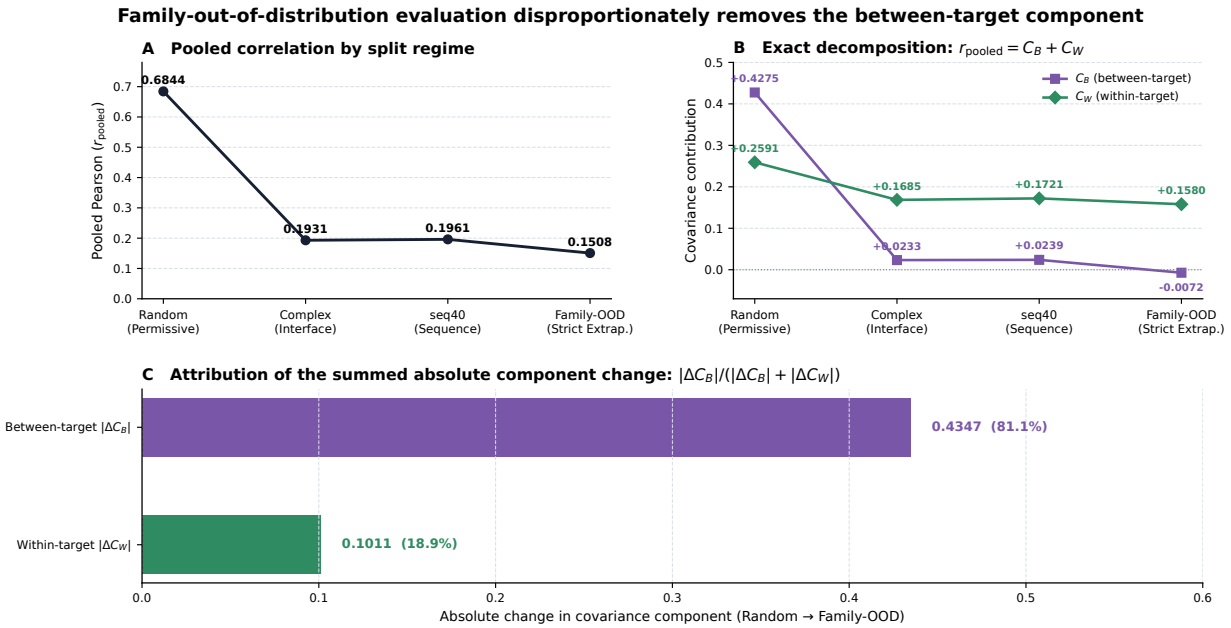


Figure 2: Family-out-of-distribution evaluation disproportionately removes the between-target component. (A) Pooled Pearson by split regime. (B) Median component summaries; the exact identity $r_{\text{pooled}} = C_{\text{B}} + C_{\text{W}}$ holds per configuration, but the plotted medians are marginal statistics computed separately and therefore need not sum exactly. C_{W} remains positive. (C) Attribution of the summed absolute component change, of which the between-target component accounts for 81.1%; the paired attribution is 81.0%. All four regimes use the same 7,523 rows with identical model specifications and seeds; complex and `seq40` are similaral and are not independent comparisons.

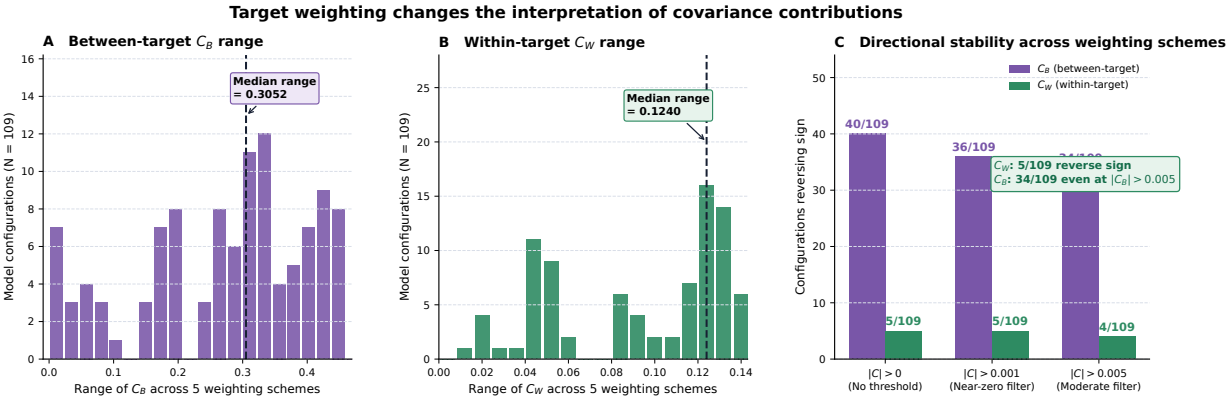


Figure 3: Target weighting changes the interpretation of covariance contributions. Predictions are unchanged; only aggregation weights differ. (A, B) Distribution of the range of C_{B} and C_{W} across the five schemes, with the median marked. (C) Configurations reversing sign at three thresholds. Under the target-total-mass weighting, C_{B} reverses sign in 40/109 configurations and still in 34/109 once $|C_{\text{B}}| > 0.005$, whereas C_{W} reverses sign in only 5/109; neither component's magnitude is invariant across schemes.

Fixed-K shortlist evaluation is constrained by candidate-set geometry

Fixed-K top-K metrics are used for mutation library or shortlist screening[11]. To isolate candidate-set geometry from the realised magnitudes of $\Delta\Delta G$, we evaluate fixed-K selection under a canonical rank-linear utility reference (Supplementary Note 4) and define the geometric discriminability surrogate

$$D(n, K) = 0 \quad (n \leq K), \qquad D(n, K) = \frac{K\,(n - K)}{n\,(n + 1)} \quad (n > K). \tag{4}$$

When a target has no more than K mutations, any model selects every candidate and the target carries no information for fixed-K shortlist metrics. In this benchmark, 53 of 130 targets with $n \geq 5$ (40.8%) fall in that degenerate regime. D is not monotonic in n: it peaks at

$$n^* = K + \sqrt{K^2 + K}, \tag{5}$$

which for $K = 10$ gives $n^* \approx 20.5$, so that $D(20) = 0.2381$ while $D(50) = 0.1569$. This surrogate is reference-dependent: only the degenerate case $n \leq K \Rightarrow D = 0$ holds for an arbitrary utility profile (full derivation, including the definitions of $E_{\max}$ and E_{rand}, in Supplementary Note 4). This geometric result does not imply that SKEMPI $\Delta\Delta G$ values are intrinsically too similar; establishing that would require an analysis using actual $\Delta\Delta G$ utility magnitudes, which we did not perform.

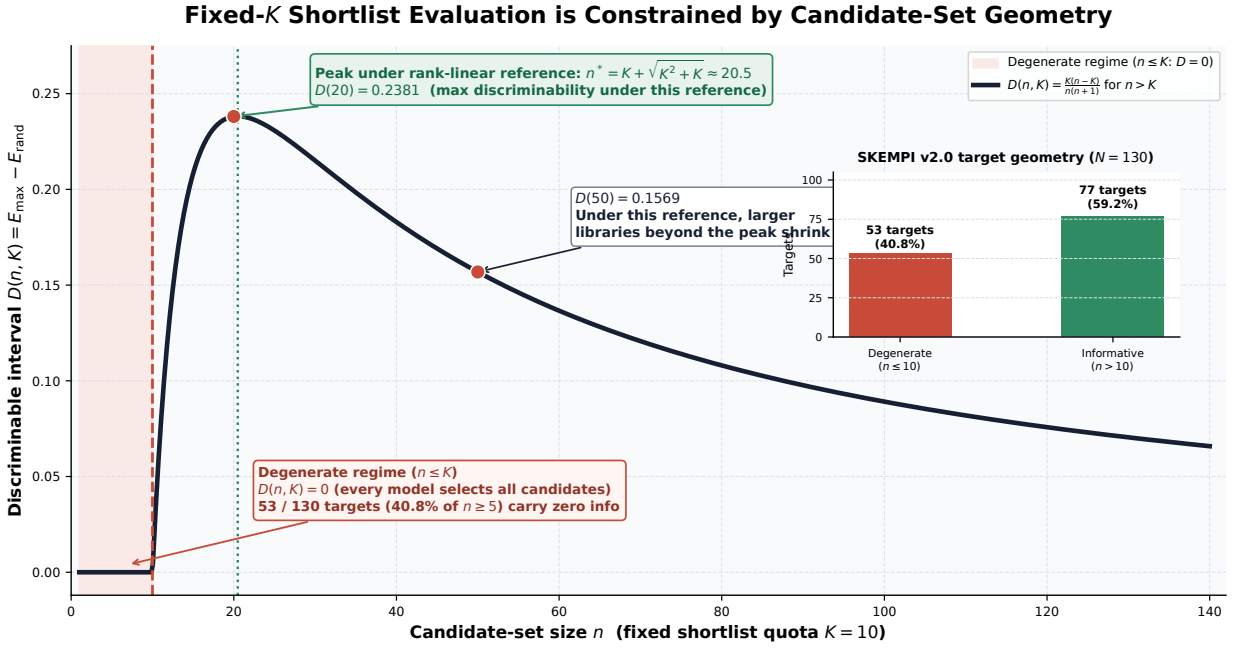


Figure 4: Fixed-K shortlist evaluation is constrained by candidate-set geometry. Under the canonical rank-linear utility reference the discriminability surrogate $D(n, K)$ is zero when $n \leq K$ and follows $K(n - K)/[n(n + 1)]$ otherwise, peaking at $n^* = K + \sqrt{K^2 + K} \approx 20.5$ for $K = 10$. In this benchmark 40.8% of targets with $n \geq 5$ lie in the degenerate regime where every model selects all candidates.

Capability decomposition clarifies retrospective remedy effects

In a matched ablation set sharing an identical 3,351-row universe, the decomposition distinguishes modifications that raise pooled scores only from those that also improve within-target ranking. Relative to a mutation-only baseline, adding static local geometry raised pooled Pearson by about 0.028 while leaving within-target Macro Spearman essentially unchanged (-0.002); mutation $\times$ partner perturbation improved both axes ($+0.070$ and $+0.044$); and a full local-physics model gave the largest improvement on both axes ($+0.105$ and $+0.076$). This analysis is retrospective and does not establish prospective prediction of which modification will succeed.

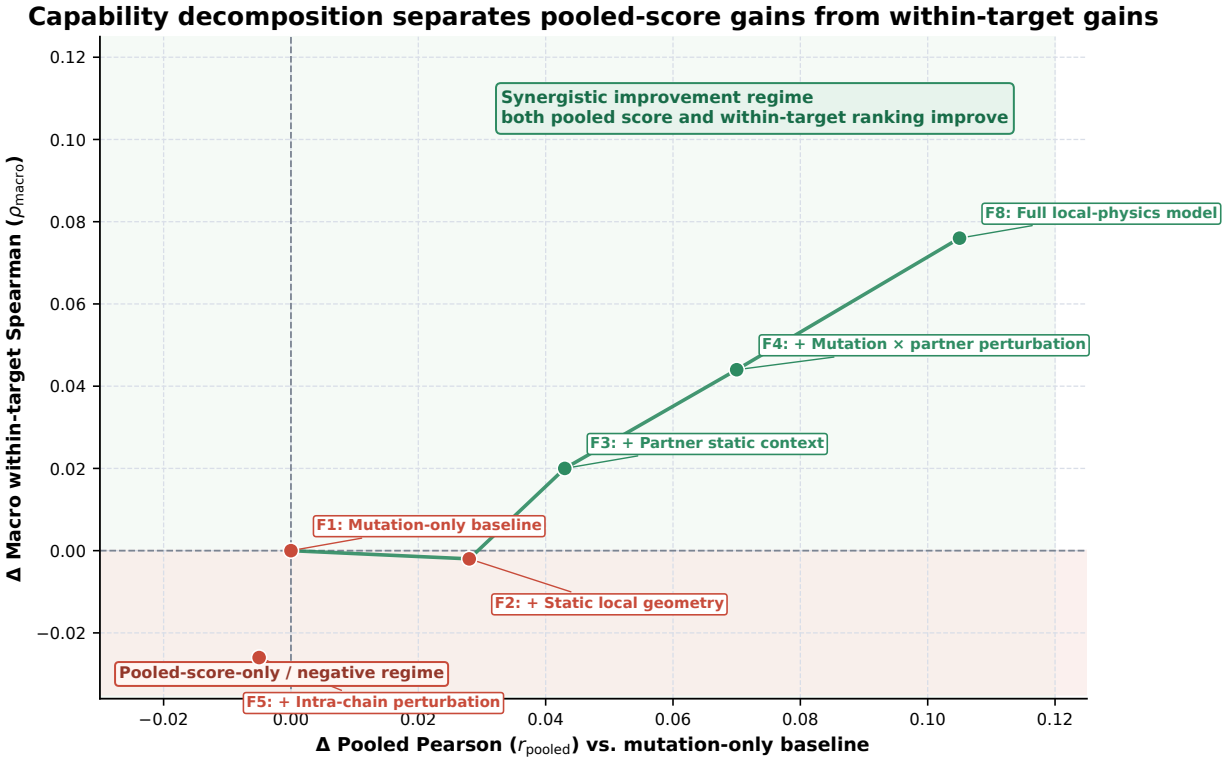


Figure 5: Capability decomposition separates pooled-score gains from within-target gains. Retrospective matched ablation on an identical 3,351-row universe; the baseline is the mutation-only model. Green markers indicate improvement on both axes; red markers indicate a pooled-score-only or negative change. The analysis is retrospective and does not imply prospective prediction of which modification will succeed.

External methods define the boundary of generalisation

We evaluated five published per-mutation prediction columns compiled in the SKEMPI-derived annotation table distributed with CATH-ddG[3] (`skempi_v2.csv`): Flex ddG[6], FoldX[7], ProteinMPNN[8], ESM-IF[9] and DDGPred[4]. Where the matching key was not unique on the annotation side, only uniquely mapped rows were used; of 4,473 rows only 1,781 have a unique mapping, and ambiguous keys were excluded rather than forcibly resolved. Each method was evaluated on its own uniquely mapped subset, and coverage therefore differs slightly across methods: Flex-ddG and DDGPred map 1,755 and 1,756 mutations (65 targets with $n \geq 5$), whereas FoldX, ESM-IF and ProteinMPNN map 1,776 mutations (66 targets with $n \geq 5$). We therefore do not claim a fully matched panel. The coverage difference is at most 21 mutations (1.2% of the panel), and we tested its influence directly: restricting all five methods to their common intersection (1,752 rows, 163 targets, 65 with $n \geq 5$) leaves the pooled ranking, the within-target ranking, the single adjacent swap and the rank agreement unchanged ($\rho = +0.90$, exact permutation $p = 0.083$; Supplementary Note 7). On the raw columns the pooled and within-target rankings are strongly positively aligned: Spearman $\rho = +0.90$ (Kendall $\tau = +0.80$, exact permutation $p = 0.083$ for $n = 5$ methods), with DDGPred first on both axes and a single adjacent swap between FoldX and Flex-ddG. This agreement is however not robust to score orientation. Because the published columns carry no documented sign convention (Methods), we also evaluated an exploratory sign-oriented panel, in which each column is aligned to the response by a single sign bit determined from the evaluation data; there the same rankings show no significant agreement ($\rho = +0.30$, $\tau = +0.20$, $p = 0.683$), because ESM-IF moves from last to second on the pooled axis while remaining last on the within-target axis. The external evidence therefore does not support a claim in either direction: with five methods, and with the orientation of the published scores undetermined by their own documentation, the panel is both underpowered and orientation-dependent, and we draw no directional conclusion from it.

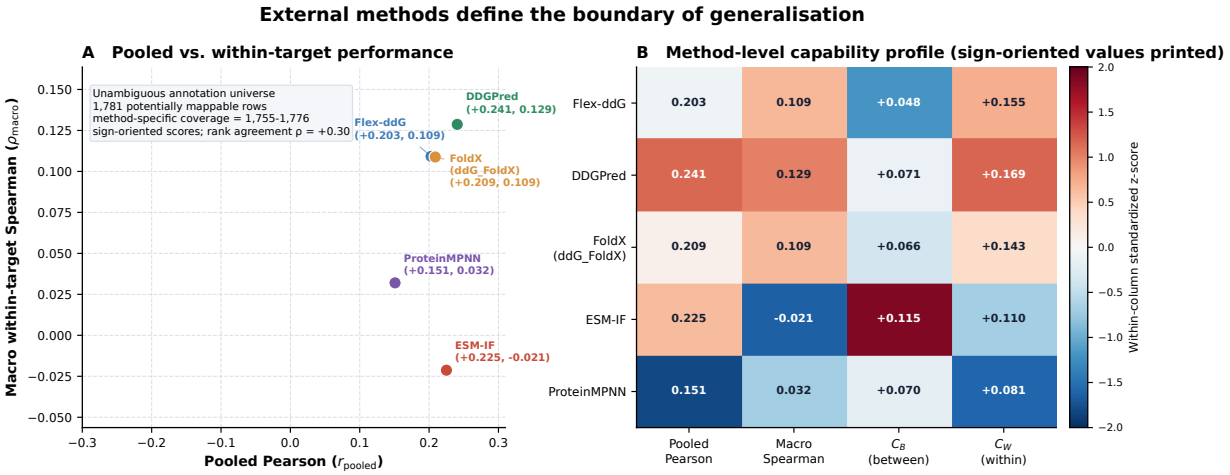


Figure 6: External methods define the boundary of generalisation. (A) Pooled versus within-target performance for five independently developed methods, each evaluated on its own uniquely mapped subset of a common unambiguous annotation universe. (B) Method-level capability profile, standardised within each metric for visual comparison only; sign-oriented values are printed in each cell. After orientation the two axes show no significant rank agreement ($\rho = +0.30$, exact permutation $p = 0.683$); used raw the same numbers would give $\rho = +0.90$ ($p = 0.083$). The panel is underpowered with five methods.

Table 3: External method capability profiles on the unambiguous mapping set.

Method	Mutations	Targets ($n \geq 5$)	Pooled Pearson	Macro Spearman	C_{B}	C_{W}	RMSE
Flex-ddG	1,755	65	0.2028	0.1092	+0.0478	+0.1550	2.1636
DDGPred	1,756	65	0.2406	0.1287	+0.0715	+0.1690	2.1262
FoldX (ddG_FoldX)	1,776	66	0.2089	0.1088	+0.0657	+0.1433	2.3691
ESM-IF†	1,776	66	+0.2250	-0.0213	+0.1149	+0.1101	—
ProteinMPNN	1,776	66	0.1509	0.0320	+0.0699	+0.0810	—

Each method is evaluated on its own uniquely mapped subset of the common annotation namespace, so mutation counts differ slightly (1,755–1,776), and their common intersection is analysed in Supplementary Note 7. Raw prediction columns are ingested and preserved exactly as published, with no rescaling or recalibration (Supplementary Note 8). A sign multiplier is applied only for the orientation sensitivity described in Methods. RMSE is reported only for methods whose output is on a $\Delta\Delta G$-like scale; ESM-IF and ProteinMPNN return dimensionless mutation-effect scores (raw standard deviations 0.093 and 0.012) and are therefore shown as "—", since an error in kcal/mol is not defined for them. †The published ESM-IF column is oriented oppositely (larger values mean stronger binding); it is sign-oriented to the common performance orientation before comparison (Methods; Supplementary Note 8).

3 Discussion

This study shows that the pooled correlation conventionally reported as a single number in PPI $\Delta\Delta G$ benchmarking is composed of two qualitatively different predictive signals: a between-target component that places different targets on the correct overall scale, and a within-target component that orders mutation effects correctly inside each target. Because our decomposition is an exact identity, 'what pooled performance is made of' becomes directly computable.

The relation to prior work should be delimited clearly. Structural leakage in PPI data and its dependence on splitting strategy were quantified systematically by Bushuiev et al., who also proposed interface-similarity splitting[2]; CATH-ddG adopted CATH homologous-superfamily splits and used structural similarity to define harder extrapolation mutations[3]; and per-complex or per-target reporting together with shortlist-style evaluation already appear in the $\Delta\Delta G$ literature[4;5;10]. Our contribution lies not in any single one of these ingredients but in the systematic integration of an exact between/within-target covariance decomposition with four audits — split strictness, target weighting, matched rank agreement and candidate-set geometry — inside one matched, provenance-filtered framework, together with a retrospective attribution case study.

The interpretation of C_{B} and C_{W} has clear boundaries. They are covariance contributions within one identity and share a common scale, so they sum to pooled correlation; they are not two independently estimated skill scores and should not be read as causal mechanisms. A model may obtain a substantial pooled correlation largely by placing targets on approximately the right overall scale while its within-target ordering ability remains limited — precisely why pooled metrics can overstate practical screening value.

The weighting-sensitivity audit yields a conclusion of direct practical relevance: the sign of the between-target component is sensitive to the aggregation weights across a wide range of magnitudes (34/109 configurations reverse sign even when $|C_{\mathrm{B}}| > 0.005$), whereas the sign of the within-target component is far more stable (5/109), although its magnitude is not. Whether between-target information appears to be present may therefore depend more strongly on the weighting convention than on the model. Any reported decomposition should declare its weighting scheme.

The candidate-set geometry result has direct implications for Top-K practice. When a target has no more than K mutations, fixed-K shortlist metrics cannot distinguish models structurally; roughly four in ten targets with $n \geq 5$ fall in this regime in our benchmark. Under the canonical rank-linear reference used here, the geometric discriminability surrogate first rises and then falls with candidate-set size, which illustrates that candidate-set size alone does not guarantee monotonically greater geometric separation under that reference. The reference-free conclusion is narrower but important: when $n \leq K$, fixed-K shortlist metrics cannot distinguish models at all, because every candidate is selected. We therefore recommend reporting n alongside K, labelling $n \leq K$ targets separately, or adopting fraction-based evaluation — recommendations that follow from the degenerate case rather than from the surrogate.

The retrospective ablation illustrates the framework's explanatory value: it distinguishes a change that raised pooled scores only from a change that also improved within-target ranking. The retrospective nature of the case study must, however, be stated explicitly: it does not establish prospective ability to predict which modifications will work.

Finally, external evidence bounds the generalisation of these conclusions. Across five independently developed methods the raw score columns suggest strong cross-axis agreement (Spearman $\rho = +0.90$), but that agreement disappears under an exploratory sign-orientation ($\rho = +0.30$). Because the published columns document no sign convention, and because the panel contains only five methods, the external evidence neither supports nor refutes a universal decoupling of the two axes; we report both orientations rather than selecting one. The panel is underpowered, so we draw no directional conclusion from it. Future work should broaden the external panel and test the relationship between the decomposed components and realised screening yield on independent prospective datasets.

4 Methods

Dataset universes

This work uses three explicitly distinct row universes; analyses are always attributed to the universe they were computed on.

Throughout this work the analysed response is the binding-affinity change

$$y \;=\; \Delta\Delta G \;=\; \Delta G_{\mathrm{mut}} - \Delta G_{\mathrm{WT}}, \qquad \Delta G = 0.592 \,\ln K_{\mathrm{D}}[\mathrm{M}], \tag{6}$$

so that $y > 0$ denotes *weaker* binding and $y < 0$ *stronger* binding. This matches the SKEMPI v2.0 $\Delta\Delta G$ convention and the convention of the external annotation table used here, so no sign inversion is applied anywhere in the pipeline. Predictions are handled in the same orientation as they are published (Methods, *External prediction semantics and units*); we note that the published ESM-IF column is oriented oppositely, carrying a negative pooled correlation, which is a property of that score rather than of our processing.

Controlled-split universe (7,523 rows). The full set of forward-paired true $\Delta\Delta G$ observations, comprising 7,523 mutation rows across 230 complexes and 112 structural families, with 172 targets having $n \geq 5$.

This universe is used only for the controlled split experiment (Table 2, Figure 2), where the same model specifications and seeds are evaluated under each split regime. It contains 4,812 distinct (complex, mutation) keys; the remaining rows are repeated experimental records of the same key (2,011 keys with more than one record, of which 1,405 carry different $\Delta\Delta G$ values).

Annotation namespace (4,473 rows). The feature-complete subset of the above universe — 4,473 rows, 214 targets, 108 structural families, 130 targets with $n \geq 5$. It is the namespace against which external predictions are mapped, and from which the primary matched universe is selected.

Primary matched universe (3,351 rows). A provenance-filtered, row-set-matched subset — 3,351 rows, 102 targets with $n \geq 5$, 10 method families — selected by a performance-blind lexicographic rule fixed before any metric values were inspected. All capability-resolved analyses, the weighting audit and the retrospective ablation are computed on this universe unless stated otherwise.

External unambiguous universe (1,781 rows). The subset of the annotation namespace for which the normalised matching key is unique (168 targets; 67 with $n \geq 5$); 2,048 ambiguous and 644 unmatched rows were excluded rather than forcibly resolved.

A target is defined as an independent binding-interface unit; structural families are assigned by sequence-identity clustering (Methods, *Structural-family definition*). Note that *structural* families (108 in the annotation namespace) are distinct from *method* families (10 in the primary matched universe), which group model variants.

Prediction sources and provenance criteria

Prediction outputs were classified by their recorded supervised training split regime. Configurations trained under a permissive (random / complex / `seq40`) partition were excluded from deployment-oriented comparisons because their evaluation rows may overlap their training folds; they were retained only in the controlled split experiment. Analyses were performed at artifact, configuration and method-family level, with one performance-blind representative per independent family.

Matched evaluation sets

Each prediction output received a SHA-256 fingerprint of its sorted row identifiers; cross-model comparison was performed only between outputs sharing a fingerprint (Supplementary Note 2, Supplementary Fig. S1). The primary matched set was selected by a rule fixed before any metric values were inspected.

Between-/within-target signal decomposition

Row-level weights are used and the identity residual is reported as a quality-control quantity. Full derivation in Supplementary Note 1.

Target-weighting schemes

Predictions are unchanged; only the target-level aggregation weight varies: per-mutation ($a_c = n_c$), per-target ($a_c = 1$), proportional to
$sqrtn_c$, proportional to
$log(1 + n_c)$, and capped at 30 ($a_c =$
$min(n_c, 30)$). The corresponding weighted decomposition, and the proof that $r_w = C_{mathrmB,w}+C_{mathrmW,w}$ holds for every scheme, are given in Supplementary Note 5.

Ranking and shortlist metrics

Within-target ordering used per-target Spearman ρ and Kendall τ correlations, pairwise accuracy and the within-target covariance contribution; shortlist metrics used fixed-K and fraction-based hit rate, precision,

recall, NDCG and enrichment factor[11].

Statistical analysis

Rank agreement uses Spearman ρ with Kendall τ as secondary; interval estimates use target-cluster bootstrap (5,000 resamples, seed 42)[12]; small external panels use exact permutation tests. All cross-metric comparisons use higher-is-better orientation[13], and every prediction column is sign-oriented to a common performance orientation before any cross-method comparison (see *External prediction semantics and units*).

Structural-family definition

Structural families group complexes whose binding interfaces are homologous. They are assigned by sequence-identity clustering of interface chains followed by union–find transitive closure: two complexes join the same family if any of their chain clusters share at least 40% sequence identity. The annotation namespace comprises 108 structural families, and family membership is frozen in the canonical target-label table used by every downstream analysis. Structural families are the unit of the family-out-of-distribution split and are unrelated to the *method* families described below.

Prediction artifact hierarchy

Prediction outputs are organised in three explicitly distinct levels, and every reported comparison states the level at which it was computed.

- **Artifact** — one model-training run together with the row set it was evaluated on, identified by a row-set fingerprint. The registry contains 12 artifacts spanning 7 distinct fingerprints.
- **Configuration** — one trained model variant within an artifact, identified by a configuration key and a random seed. The full registry holds 202 configurations.
- **Method family** — a group of configurations that share an architectural or feature-design lineage (for example the ten ablation families F0–F9 of the local-physics model). Families, not individual configurations, are the unit of independent evidence, because configurations within a family are not independent observations.

Cross-model comparison is only ever performed between configurations whose row-set fingerprints are identical; metrics are never compared across fingerprints. The primary matched universe contains 30 configurations drawn from 10 method families (three seeds per family, with F0/F1/F2/F3/F4/F5/F6/F7/F8/F9 lineage labels).

Training-split provenance and known-overlap filtering

Each configuration records its supervised training split regime as one of *random*, *complex*, *seq40* or *family-OOD*. Configurations trained under a permissive regime (random / complex / `seq40`) are excluded from all deployment-oriented comparisons, because their evaluation rows may overlap their training folds; they are retained only in the controlled split experiment, where the split regime is itself the variable under study. Records of exclusions are retained in the leakage-exclusion log shipped with the analysis. Throughout, we claim control only of *task-specific supervised* overlap; we cannot exclude the possibility that foundation-model pretraining encountered similar sequences or structures.

Row-set fingerprinting

Each prediction output receives a SHA-256 fingerprint computed over its sorted evaluation row identifiers. The fingerprint is the sole basis for deciding that two configurations are directly comparable, which makes the matching rule auditable and independent of any metric value. Seven distinct fingerprints occur across

the artifacts used here; Table S1 lists each stratum with its fingerprint, row count, target and family counts, and the number of models available on it.

Primary matched universe selection

The primary matched universe was selected by a performance-blind lexicographic rule fixed before any metric values were inspected, ordering candidate strata by (1) provenance cleanliness, (2) number of independent method families, (3) number of targets with $n \geq 5$, and (4) number of rows. This yields the `f8c33c7e` stratum: 3,351 rows, 102 targets with $n \geq 5$, and 10 method families. Because the rule never references a performance metric, the selection cannot be tuned towards a favourable result.

Representative selection within method families

Analyses that require one entry per independent lineage use a single *performance-blind representative* per method family, chosen by configuration identifier rather than by score. This rule is recorded per configuration in the model-family registry, together with its stated reason. Reporting at the family level avoids treating seed replicates as independent evidence, which would otherwise inflate the apparent number of independent methods.

Score orientation and its limits

The five external score columns are distributed as part of a published annotation table and we were unable to locate, in the accompanying repository or documentation, a statement of the score definition or sign convention used to produce them. We therefore treat the published columns as *raw* and make the raw analysis the primary external analysis: it requires no orientation decision and is fully provenance-driven.

Because a score whose sign convention is opposite to the response would be penalised for its convention rather than for its predictions, we additionally report an *exploratory* sign-orientation sensitivity. Writing

$$s_m^{\text{aligned}} = o_m \, s_m^{\text{raw}}, \qquad o_m \in \{+1, -1\}, \tag{7}$$

we determine o_m from the sign of the within-complex correlation between column m and the evaluation response $y = \Delta\Delta G$ (which removes between-target differences and so isolates direction rather than predictive skill). This gives $o_m = -1$ for ESM-IF (within-complex correlation -0.24) and $o_m = +1$ for Flex-ddG, FoldX, DDGPred and proteinMPNN ($+0.49$, $+0.49$, $+0.51$, $+0.30$).

We state the limitation of this procedure explicitly: **the orientation multiplier is selected using the evaluation labels**, because no label-independent source for it could be found. It is a single sign bit per method, it involves no rescaling or recalibration, and it preserves all information content; but it is a label-informed decision and we therefore report it as a sensitivity analysis rather than as the primary result. The raw columns are preserved unmodified in Supplementary Table S3, which also records o_m. The two orientations lead to opposite conclusions about cross-axis agreement, and we report both.

External prediction mapping and ambiguity handling

Two distinct failure modes are handled separately and are never conflated. A prediction is *unmatched* when its normalised matching key has no counterpart on the annotation side; it is *ambiguous* when the key maps to more than one annotation row. Of the 4,473 annotation rows, 644 are unmatched and 2,048 are ambiguous, leaving 1,781 uniquely mapped rows. Ambiguous keys are excluded rather than forcibly resolved by an arbitrary tie-break, because any deterministic tie-break would inject an unquantified bias. Each external method is therefore evaluated on its own uniquely mapped subset, and coverage differs slightly across methods (1,755–1,776 mutations).

External prediction semantics and units

The five external methods originate from a single published annotation table of SKEMPI-derived protein–protein interaction mutations, and each prediction column is ingested and preserved exactly as published, with no rescaling and no recalibration; the raw values are reproduced directly from the published columns to four decimals for all five methods (Supplementary Note 8). A sign multiplier is applied only for the exploratory orientation sensitivity described below, and never to the raw columns themselves. Three columns (Flex-ddG, FoldX and DDGPred) are quantitative $\Delta\Delta G$ predictions on a kcal/mol-like scale, whereas ESM-IF and ProteinMPNN are inverse-folding sequence models whose published outputs are dimensionless mutation-effect scores rather than energies. We therefore report RMSE only for the first group; for the latter two an error in kcal/mol is not defined and is shown as "—". Correlation-based quantities — pooled Pearson, macro within-target Spearman, C_{B} and C_{W} — are invariant to positive affine rescaling (a positive change of scale and origin) and are reported for all five methods; a negative scaling would reverse their orientation and sign.

Bootstrap and permutation inference

Interval estimates for target-level quantities use a cluster bootstrap that resamples targets, not rows, so that within-target dependence is preserved; 5,000 resamples are used throughout with a fixed seed. For the controlled split, intervals are computed by resampling the matched configuration pairs. For the five-method external panel, the exact permutation distribution of Spearman's ρ is enumerated over all $5! = 120$ relabellings, giving $p = 0.083$ for the raw columns and $p = 0.683$ after the exploratory sign-orientation; asymptotic approximations are not used because the panel is far too small for them to be reliable.

Software and reproducibility

Analyses were run under Python 3.11 with `numpy` 2.4, `pandas` 3.0, `scipy` 1.17, `scikit-learn` 1.9 and `matplotlib` 3.11. No new model was trained for this study and no GPU was used: all capability analyses re-analyse previously produced prediction artifacts. Random seeds are fixed and recorded per configuration (42/43/44 for the controlled split experiment; seed 42 for bootstrap resampling). Every headline number is generated from a single machine-readable results file, and an automated consistency checker re-derives the reported values from the underlying artifacts; the checker, the frozen input manifest with hashes, and the machine-readable sources are provided with the Supplementary Information.

Where the normalised matching key was not unique on the annotation side, only uniquely mapped rows were used; ambiguous keys were excluded rather than resolved.

Reproducibility and consistency validation

All headline numbers are generated from a single machine-readable file and verified by an automated consistency checker. No new model was trained and no GPU was used.

5 Limitations

Our extrapolation setting is based on target-family partitioning, which does not exclude the possibility that foundation-model pretraining encountered sequences or structures similar to the evaluation families; we can control task-specific supervised overlap only. The model pool is not a random sample of all PPI $\Delta\Delta G$ methods, and only five methods are externally evaluable, so method-level inference is underpowered.

Shortlist and candidate-selection analyses rest on re-analysis of existing predictions and include no prospective wet-lab validation; when a target has no more than K mutations the relevant metrics carry no information. The retrospective ablation likewise does not establish prospective prediction.

C_{B} and C_{W} are covariance-signal contributions rather than causal mechanisms. We performed no identity-memorisation or target-shuffling intervention and therefore make no causal claim about the origin of the between-target component.

No independent prospective experimental dataset was available, so no external dataset validation was performed. This work does not provide a universal model recommender and does not claim general superiority of any method family.

Data and code availability

All analyses use existing predictions and published external predictions; no new model was trained and no GPU was used. Frozen input hashes, the machine-readable source of every headline number and an automated consistency checker are provided with the Supplementary Information and with the manuscript. A public repository will be linked here upon acceptance.

Author contributions

Study conception, analysis design, interpretation of results and writing of the manuscript were performed by the author(s). The use of AI-assisted tools is declared below.

Use of AI-assisted tools

AI-assisted coding and language tools were used to support code drafting and debugging, generation of plotting scripts, document automation, literature and repository retrieval, and language editing. All metric definitions, analysis choices, scientific interpretations, hypothesis decisions and final manuscript content were reviewed and approved by the authors, who take full responsibility for the work. AI tools were not used to generate experimental data or to modify upstream model predictions.

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